StoryForge — Complete Master Stack

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Overview

StoryForge is one platform with three products: prompt to script to video, text or link to animation, and AR/VR/MR/XR experiences. It runs on six dashboards over one shared pipeline, AI layer and big-data layer.

AI engine decision: hybrid. Launch on third-party AI APIs behind an in-house model router, then move high-volume work to self-hosted open models. This protects the platform when a vendor shuts down a model, lets each job use the best model for it, and lowers cost once volume reaches hundreds of clips per day.

Design rule: everything is asynchronous. Every generation step is a job on a queue with retries, because video models return results minutes later, not instantly.

Scale target: investor-scale. The design assumes millions of users, multi-region delivery and a full big-data stack from the start.

Dashboards

#	Dashboard	What the user does	Output
1	Script-to-Video Studio	Writes a prompt, approves the script, reviews the video up to 5 times	Finished video in the chosen language, ready to publish
2	Animation Studio	Enters text or pastes a link to content they own, picks a voice	Animated video with the selected voice
3	AR Studio	Builds or generates 3D scenes placed in the real world	AR experience for phones and glasses
4	VR Studio	Builds or generates fully immersive scenes	VR experience for headsets
5	MR/XR Studio	Blends real and virtual content, multi-user sessions	Mixed-reality experience across devices

#	Dashboard	What the user does	Output
6	Admin and Analytics	Staff monitor users, revenue, GPU cost, queues, moderation	Operations control and investor metrics

Generation pipeline

Every video passes through these steps in order. Temporal runs the workflow and pauses at the two human steps (script approval and review).

1. **Intake.** User writes a prompt, or pastes a link in the Animation Studio. Link intake downloads the source, transcribes it and detects scenes, after a rights confirmation.
2. **Prompt moderation.** A safety filter blocks banned content before any money is spent on generation.
3. **Cost estimate.** The system shows the credit cost and checks the plan's budget cap.
4. **Script generation.** The LLM writes the script and a scene list.
5. **User approval.** The user approves, edits or regenerates the script.
6. **Localization.** The script is adapted per target language with a glossary; top languages go to a native-speaker review queue.
7. **Scene planning.** Each scene gets a visual prompt, style profile, reference images and fixed seed for character consistency.
8. **Parallel generation.** Voice (TTS), visuals (video or animation models), avatars with lip-sync, and music and sound effects are generated at the same time.
9. **Assembly.** FFmpeg workers stitch scenes, level the audio, and add subtitles from forced alignment (SRT/VTT or burned-in).
10. **Automated quality check.** A vision model scores every clip and retries failures before the user sees anything.
11. **Output moderation and provenance.** A final safety check, then an AI-content watermark and C2PA label.
12. **User review.** Up to 5 revision rounds, limited by plan. The timeline editor regenerates a single scene, not the whole video.
13. **Export and publish.** Download, or direct scheduled upload to YouTube, TikTok, Facebook and Instagram.
14. **Notify.** Email, push and in-app alerts at script-ready, video-ready and publish.

Core platform stack

Layer	Stack	Job
Web frontend	Next.js, TypeScript, Tailwind, WebSockets	All six dashboards, live progress, review player, timeline editor
Mobile app	React Native or Flutter	Review, approval, notifications
API gateway	Kong or Envoy	Routing, rate limits, plan enforcement
Auth	Keycloak or Auth0	Login, SSO, roles, workspaces
AI services	Python, FastAPI	Script, voice, video, QC, moderation services
Business services	Go or NestJS	Accounts, billing, credits, teams, notifications
Workflow engine	Temporal	Long-running pipeline with retries and human pauses
Event backbone	Apache Kafka	Every job, click and render event
Job queues	Kafka plus priority queues per plan	Paid plans render first
Rendering	FFmpeg workers on GPU node pools	Stitching, subtitles, audio, export
Orchestration	Kubernetes, KEDA autoscaling	Scales workers and GPUs with queue depth
Object storage	S3-compatible storage plus CDN	Video files, assets, 3D models
Core database	PostgreSQL	Users, projects, plans, revisions, credit ledger
Cache	Redis	Sessions, rate limits, job status
Vector database	pgvector or Qdrant	Asset and style search
Search	OpenSearch	Projects, scripts, templates, assets
Notifications	Email service, push service, in-app feed	Ready and failure alerts

AI layer

All models sit behind one model router that picks a provider per job by quality, price, language and availability, and fails over automatically.

Capability	Launch (APIs)	Later (self-hosted)
Script and scene writing	LLM APIs	Same, with prompt library and evaluation sets
Translation and localization	LLM plus per-language glossary	Same, plus native-speaker review queue
Voice (TTS)	ElevenLabs, Google Cloud TTS, Azure Speech	Open TTS models for high-volume languages
Voice cloning	Provider cloning with consent check	Self-hosted cloning
Video generation	Veo, Kling, Seedance, Runway through an aggregator	Wan and LTX families on own GPUs
Animation	Image and video models with style profiles	Fine-tuned open models per style
Avatars and lip-sync	Talking-avatar API	Open lip-sync models
Music and sound effects	Licensed stock library or music-generation API	Same
Subtitles	Whisper-style transcription and forced alignment	Self-hosted Whisper
Character consistency	Reference images, style profiles, fixed seeds	LoRA fine-tunes per character
Quality check	Vision model scoring each clip	Same, tuned on user rejection data
Moderation	Text, image, video and audio safety classifiers	Same
3D generation	Text-to-3D and image-to-3D APIs	Open 3D models

ML operations: model registry, evaluation sets, A/B tests comparing models on quality and cost, prompt versioning, and a feature store for recommendations. Model serving for self-hosted models runs on Triton or vLLM-style servers with GPU autoscaling.

Big data layer

Every event in the platform lands in one lakehouse, which feeds analytics, cost tracking, recommendations and model improvement.

Stage	Stack	Job
Ingestion	Kafka, Kafka Connect, Debezium (database change capture)	Streams events and database changes
Stream processing	Apache Flink	Real-time usage, queue health, fraud signals, live cost
Batch processing	Apache Spark	Daily aggregates, training datasets, heavy joins
Lakehouse	Apache Iceberg on object storage	Single store for raw and cleaned data
Transformation	dbt	Clean, tested analytics tables
Orchestration	Apache Airflow	Schedules batch jobs and data pipelines
Fast analytics	ClickHouse	Sub-second dashboards for admin and users
BI	Superset or Metabase	Internal reports and investor metrics
Governance	Data catalog, quality checks (Great Expectations), lineage	Trustworthy, traceable data
ML data	Feature store, labeled QC and rejection data	Recommendations and model tuning

Key outputs: cost per video minute, gross margin per plan, model quality scores, revision rates per language, retention and churn.

XR layer

The AR, VR and MR/XR dashboards share one 3D asset pipeline and differ only in how scenes are delivered.

Part	Stack	Job
Engine	Unity with OpenXR	One codebase for most headsets
Mobile AR	AR Foundation (ARKit and ARCore)	AR on iOS and Android
Browser XR	WebXR with Three.js or Babylon.js	No-install previews and sharing
3D asset generation	Text-to-3D and image-to-3D models	Objects, characters, environments
Asset processing	Mesh clean-up, retopology, texture compression, level-of-detail	Assets light enough for phones and headsets
Asset formats	glTF and USD	Standard delivery and editing formats
Scene editor	In-browser 3D editor	Users arrange generated assets
Spatial audio	3D audio engine with the generated voice placed in space	Immersive narration
Multi-user	Real-time networking (Photon or Colyseus)	Shared sessions in MR/XR
Cloud streaming	Pixel streaming from GPU servers	Heavy scenes on weak devices
Device targets	Meta Quest, Apple Vision Pro, Android, iOS, web	Export and store packaging

Platform features

Feature	What it includes
Teams and workspaces	Roles, shared projects, approval flows
Brand kits	Logo, fonts, colors, intro and outro, default voice

Feature	What it includes
Templates	Ready formats: documentary, shorts, explainer, ad, story
Asset library	User uploads, stock footage, music, saved characters and styles
Timeline editor	Scene-level edits, regenerate one scene, swap voice or music
Publishing	Direct upload and scheduling to YouTube, TikTok, Facebook, Instagram
Notifications	Email, push, in-app
Public API and webhooks	Agencies and developers generate videos programmatically
Mobile app	Review, approve, publish on the go
Support	Help center, ticketing, in-app chat, automatic credit return for failed renders
Search	Across projects, scripts, templates and assets
Version history	Every script and video revision kept and restorable

Infrastructure, security and compliance

Area	What to build
Cloud	AWS or GCP as primary, infrastructure as code with Terraform
Deployment	ArgoCD (GitOps), staged rollouts, feature flags
Monitoring	Prometheus, Grafana, OpenTelemetry tracing, central logging, on-call alerts
Cost control	Per-user and per-plan budget caps, cost estimate before generation, GPU spot instances, cost alerts
Fraud and abuse	Bot detection, free-tier limits, stolen-card checks, device fingerprinting

Area	What to build
Security	Secrets vault, encryption at rest and in transit, web firewall, DDoS protection, audit logs, penetration tests
Compliance	GDPR-style data export and deletion, consent records, SOC 2 readiness
AI provenance	Watermarking and C2PA labels on every output
Voice and likeness	Consent verification before cloning any voice or face
Multi-region	Regional storage and CDN, database replicas, failover
Disaster recovery	Automated backups, restore drills, defined recovery targets
Testing	Unit, integration and load tests; pipeline tests with cheap test models

Business layer

Area	What to build
Plans	Three tiers (for example Starter, Pro, Studio), each with video minutes, revision rounds, languages, resolution and queue priority
Credits	Credit ledger in PostgreSQL; every generation debits credits, failed renders refund automatically
Payments	Stripe where available, plus a merchant-of-record service for tax and unsupported countries
Regional pricing	Local currency and price levels per market
Legal	Terms of service, content-ownership rules, rights confirmation for links, takedown process, acceptable-use policy

Area	What to build
Growth	Affiliate and referral program, free trial credits, creator templates marketplace
Investor metrics	Cost per video minute, gross margin per plan, retention, churn, revenue per user

Plan prices and exact limits are still open and should be set from the measured cost per video minute.

Build order

Each phase ships something usable and the later phases reuse the earlier ones.

1. **Foundation.** Auth, workspaces, plans, credit ledger, Kafka, Temporal, storage, CI/CD, monitoring.
2. **Script-to-Video Studio.** Full pipeline from prompt to reviewed video, with moderation, cost estimate, QC and 5 revisions. This is the investor demo.
3. **Localization and publishing.** Multi-language voices, subtitles, direct upload to social platforms, notifications.
4. **Big data layer.** Events into the lakehouse, cost and usage dashboards, Admin and Analytics dashboard.
5. **Animation Studio.** Link intake, style profiles, character consistency, avatars and lip-sync.
6. **Platform features.** Teams, brand kits, templates, public API, mobile app, support.
7. **AR Studio**, then **VR Studio**, then **MR/XR Studio**, on one shared 3D asset pipeline.
8. **Self-hosted GPU cluster.** Move the highest-volume models in-house, guided by cost data from phase 4.
9. **Scale-out.** Multi-region, compliance certification, marketplace.

Second-pass additions

These 24 items were absent from the sections above and complete the system.

Area	Component	What it adds
Content	Research and fact-check step	Web research with cited sources before script writing, so documentary scripts are accurate
Content	Title, thumbnail, description and tags generator	Publish-ready packaging with each video, including thumbnail image generation
Content	Image generation and stock footage	Image models plus stock libraries (Pexels, Storyblocks) for scenes that need no video model
Content	Multi-format export	16:9, 9:16 and 1:1 versions, with automatic reframing for Shorts, Reels and TikTok
Content	Script structure presets	Hook, retention beats and call-to-action patterns per format and platform
Media	Video delivery	Transcoding ladder with HLS/DASH adaptive streaming, thumbnails and preview clips
Media	Upload handling	Resumable uploads, virus scanning, file-type validation
Media	Storage lifecycle	Retention rules per plan, cold storage for old projects, automatic clean-up of failed renders
Rights	Copyright detection	Fingerprint matching on uploads and links to catch protected film, music and footage
Rights	Licence tracking	Record of which music, stock and model licence covers each output
Safety	Human trust and safety queue	Reviewers for flagged content, with an appeals process

Area	Component	What it adds
Safety	Real-person and age rules	Block deepfakes of real people without consent; age gating and child-safety checks
Product	Interface languages	Dashboard translated into major languages, with right-to-left support for Urdu and Arabic
Product	Accessibility	Keyboard navigation, screen-reader support, caption defaults
Product	Real-time collaboration	Timestamped comments on videos, shared editing, approval links for clients
Product	Onboarding and product analytics	Guided first video, funnel tracking (PostHog or Amplitude), experiment platform for A/B tests
Product	Customer messaging	CRM, lifecycle email, in-app announcements, feedback and feature-request board
Enterprise	Enterprise tier	SSO and SCIM, white-label option, private deployment, dedicated capacity, SLA
Engineering	Developer platform	Monorepo, API documentation, SDKs, Kafka schema registry, service mesh
Engineering	Vendor capacity	Reserved quotas and rate-limit management with each AI provider, automatic failover
Engineering	Reliability	Public status page, incident process, defined uptime targets
Engineering	Internal admin tools	User impersonation for support, manual refunds, job replay, feature-flag console
Company	Hiring plan	Teams for backend, ML, data, XR, frontend, design, trust and safety, support

Area	Component	What it adds
Company	Budget model	Monthly cost forecast for APIs, GPUs, storage, CDN and staff, tied to user growth

Services

The platform splits into 22 services, each owning its own data and talking to the others through Kafka events and the gateway.

Service	Owns	Main endpoints or events
Auth	Users, sessions, SSO	Sign-up, login, token refresh
Workspace	Teams, roles, brand kits	Create workspace, invite member, set role
Project	Projects, versions, comments	Create project, list versions, add comment
Billing	Plans, subscriptions, invoices	Subscribe, change plan, payment webhooks
Credit ledger	Credit balances and transactions	Reserve, debit, refund credits
Moderation	Safety decisions, review queue	Check prompt, check output, appeal
Research	Sources and fact notes	Research topic, return cited notes
Script	Scripts and scene lists	Generate, edit, approve script
Localization	Translations, glossaries	Translate script, submit for review
Scene planner	Visual prompts, style profiles	Plan scenes, lock character references
Model router	Provider catalog, routing rules, quotas	Route job, report cost, fail over
Voice	TTS jobs, voice library, clones	Generate narration, list voices
Visual	Video, image and animation jobs	Generate clip, regenerate scene

Service	Owns	Main endpoints or events
Avatar	Avatar and lip-sync jobs	Generate presenter clip
Audio	Music, sound effects, mixing	Select track, level audio
Assembly	Render jobs, subtitles, formats	Render video, export format
Quality check	Clip scores, retry decisions	Score clip, request retry
Review	Revision rounds and feedback	Submit feedback, count revisions
Publishing	Social accounts, schedules, packaging	Connect channel, schedule upload
Ingestion	Link downloads, transcripts, rights records	Ingest link, confirm rights
XR	3D assets, scenes, sessions	Generate asset, build scene, start session
Notification	Templates, delivery log	Send email, push, in-app

Database schema

PostgreSQL holds these core tables; every table also carries an id, created and updated timestamps, and a workspace id for tenant isolation.

Table	Key fields	Notes
users	email, name, locale, status	One row per person
workspaces	name, plan_id, owner_id, region	Tenant boundary
memberships	user_id, workspace_id, role	Owner, admin, editor, viewer
plans	name, monthly_credits, max_revisions, max_resolution, queue_priority	Three tiers
subscriptions	workspace_id, plan_id, status, renews_at	Linked to the payment provider

Table	Key fields	Notes
credit_transactions	workspace_id, job_id, amount, type, balance_after	Append-only ledger: reserve, debit, refund
projects	workspace_id, dashboard_type, title, status, language	One per video or experience
scripts	project_id, version, content, status, approved_by	Draft, approved, rejected
scenes	script_id, position, narration, visual_prompt, style_profile_id, seed	One row per scene
jobs	project_id, scene_id, type, provider, model, status, cost, attempts	Every generation step
assets	workspace_id, type, storage_url, duration, licence_id	Clips, audio, images, 3D models
renders	project_id, version, format, storage_url, qc_score	Each assembled video
revisions	project_id, round_number, feedback, scene_ids	Enforces the 5-round limit
voices	provider, language, gender, style, is_clone, consent_record	Voice library
style_profiles	workspace_id, name, reference_assets, settings	Character and look consistency
moderation_events	subject_type, subject_id, decision, reason, reviewer_id	Audit trail and appeals
rights_records	project_id, source_url, declaration, verified	For pasted links
publish_targets	workspace_id, platform, account_ref, token_ref	Tokens kept in the vault
publications	render_id, target_id, scheduled_at, status, external_url	Upload history
xr_scenes	project_id, mode, scene_graph_url, device_targets	AR, VR or MR

Table	Key fields	Notes
audit_logs	actor_id, action, subject, ip, timestamp	Security and compliance

High-volume data (events, job logs, analytics) lives in the lakehouse and ClickHouse, not in PostgreSQL.

Kafka topics

Every topic uses a registered schema, is keyed by project or workspace id so related events stay in order, and has a matching dead-letter topic for failed messages.

Topic	Produced by	Consumed by
project.events	Project	Analytics, notification
script.requested / script.generated / script.approved	Script	Workflow, localization, analytics
moderation.decisions	Moderation	Workflow, review queue, analytics
localization.completed	Localization	Scene planner
scene.planned	Scene planner	Voice, visual, avatar, audio
job.requested / job.started / job.completed / job.failed	Model router and workers	Workflow, credit ledger, quality check, analytics
qc.scored	Quality check	Workflow, model router (quality feedback)
render.completed	Assembly	Review, notification, publishing
review.submitted	Review	Workflow, analytics
publish.scheduled / publish.completed	Publishing	Notification, analytics
credits.reserved / credits.debited / credits.refunded	Credit ledger	Billing, analytics, fraud

Topic	Produced by	Consumed by
billing.events	Billing	Credit ledger, notification
provider.health	Model router	Routing rules, alerts
xr.asset.generated / xr.session.events	XR	Analytics, notification
user.activity	Frontend and gateway	Flink, fraud, product analytics
audit.events	All services	Audit store, security monitoring

Flink reads job, credit and activity topics for live cost and fraud signals. Spark reads everything from the lakehouse for daily aggregates.

Plan limits (draft)

These limits are a starting proposal, not final numbers. Prices are left open until the cost per video minute is measured in phase 2.

Limit	Starter	Pro	Studio
Video minutes per month	20	100	500
Revision rounds per video	2	5	5
Maximum video length	3 minutes	15 minutes	60 minutes
Resolution	720p	1080p	4K
Languages	10	30	All supported
Voice cloning	No	1 voice	10 voices
Animation Studio	Limited	Yes	Yes
AR / VR / MR Studios	No	Preview	Full
Team seats	1	5	25
Queue priority	Standard	High	Highest
Direct publishing	1 channel	5 channels	Unlimited

Limit	Starter	Pro	Studio
API access	No	No	Yes
Watermark	StoryForge mark	None	None

Known limits and risks

- **Language coverage.** Native-quality voice in every language is not achievable today. Leading TTS providers cover roughly 30 to 75 languages, and quality drops outside the top ten or so. Launch with 30 to 50 strong languages and label the rest as beta.
- **Movie links.** Turning someone else's film into animation without a licence is a legal risk for the platform. Limit the feature to content the user owns or has rights to, with rights confirmation and takedown handling.
- **Unit cost.** One video can cost more than a cheap plan earns. Budget caps, cost estimates and the QC retry limit must exist from phase 1.
- **Vendor dependence.** Providers retire models with short notice. The model router and at least two providers per capability are required, not optional.
- **Team size.** This scope needs a funded team across backend, ML, data, XR and frontend. A solo build should stop at phase 2 and use it to raise funding.

Sources

- Best AI Video Generation API 2026 (Aireiter)
- Top 5 AI Video APIs Compared (Atlas Cloud)
- Best Open-Source AI Video Generation Models (Thunder Compute)
- Best Text-to-Speech APIs (Eden AI)
- Multilingual TTS: Choosing a Voice Model (Simba)